FULL SYSTEM ELECTRICAL SCHEMATIC — REVISION B

Dedicated regulated 5.0–5.1 V / 3 A USB-C source only. No USB PD request and no current-advertisement detector.

All off-board interconnects: ../Wiring/NODI_Wiring_RevB.pdf. All individual pin nets: ../netlist.csv.

EXTERNAL INSULATED INPUT POD / PURCHASED MODULE

J1
SparkFun BOB-15100 current Rd revision

CC2 1 USB_CC2
D+ 2
D- 3
CC1 4 USB_CC1
GND 5 GND
VBUS 6 +5V_RAW

USB_CC1 → R7 5.1k 1% included → GND
USB_CC2 → R8 5.1k 1% included → GND

R7/R8 shown to define the sink: they are inside J1 purchase.

F1
2.5 A time-delay >=32 VDC
1 2

D1
1N5819
A 2 → 1 VSYS_EXT

+5V_RAW → +5V_FUSED

C1
1000uF / 10V
1 → +5V_FUSED
2 → GND

C2
100nF / >=10V
1 → +5V_FUSED
2 → GND

TP1
+5V solder star
+5V_FUSED

TP2
GND solder star
GND

+5V_FUSED
VSYS_EXT
GND

D1 and Pico onboard USB diode OR the sources. External +5V must not connect to U1 pin40 VBUS.

Actuators, buffer and piezo driver

File: Motion_Audio.kicad_sch

Controller, display, RTC and button

File: Controller_Interfaces.kicad_sch

All module external pins shown; vendor PCB internals referenced separately
SYSTEM SCHEMATIC / PROTOTYPE / HARDWARE UNTESTED
NODI — David Borsky

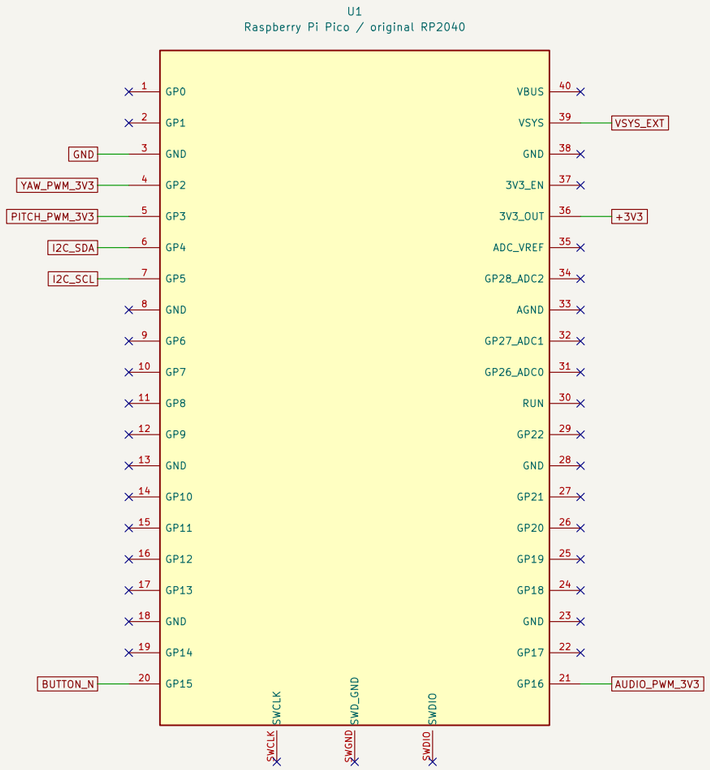
Sheet: /
File: NODI_System_RevB.kicad_sch

Title: NODI / power entry and distribution

Size: A3	Date: 2026-10-25	**Rev: B**
KiCad E.D.A. 10.0.7		Id: 1/3

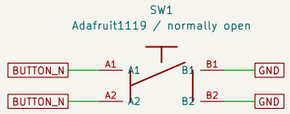
CONTROLLER AND MODULE EXTERNAL PINS

NC crosses mean no EXTERNAL wire. Purchased modules contain onboard connections and pullups not expanded here.



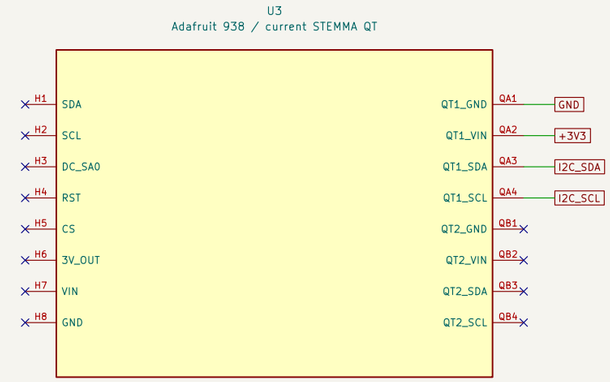
U1 programming micro-USB is ONBOARD; use its normal data cable.

U1 pin40 VBUS left externally NC. SWD pads unused.



GP15 internal INPUT_PULLUP. Wire one A terminal and one B terminal;

the second terminal of each pair is internally common, not another cable wire.



Use QT1 only: pin1 GND,2 VIN,3 SDA,4 SCL. Header H1..H8 and QT2 unused externally.

Default I2C solder jumpers closed. RST auto-reset; DC/SA0 and CS need no external wire.

U3 address0x3D or0x3C; firmware probes both. Do not connect H6 regulator output to VIN.



BT1: non-rechargeable CR1220 inserted in U4 onboard holder; no loose battery wires.

BAT,32K,SQW,RST pads left externally NC. Module includes 10k I2C pullups to VIN.

All module external pins shown; vendor PCB internals referenced separately
SYSTEM SCHEMATIC / PROTOTYPE / HARDWARE UNTESTED
NODI – David Borsky

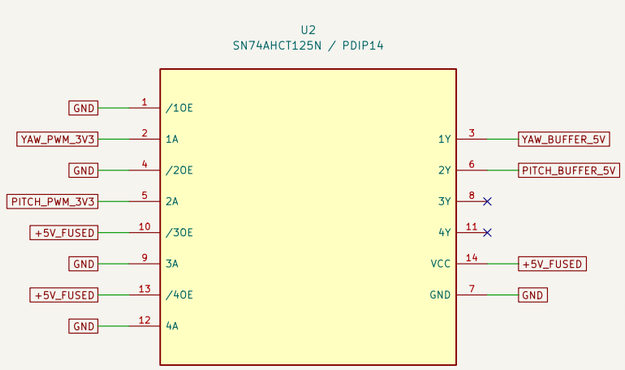
Sheet: /Controller, display, RTC and button/
File: Controller_Interfaces.kicad_sch

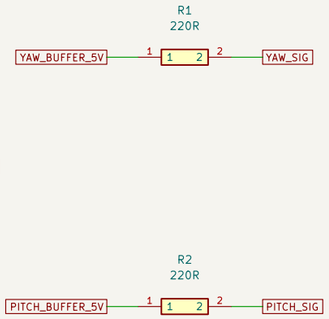
Title: NODI / controller and user interfaces

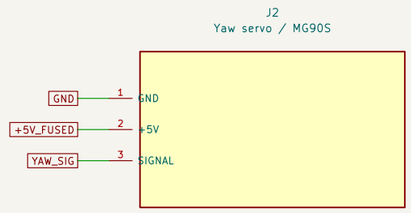
Size: A3	Date: 2026-10-25	**Rev: B**
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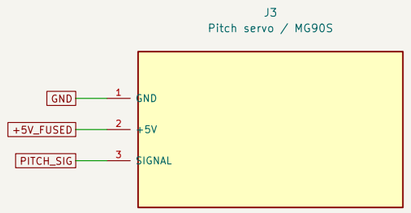
SERVO BUFFER, ALL TIES, AND DISCRETE AUDIO DRIVER

GPIO numbers and physical U1 pins match Firmware/NODI/NODI.ino. Hardware, rail droop and startup remain untested.

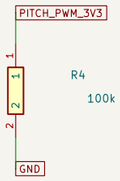


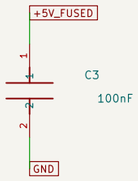


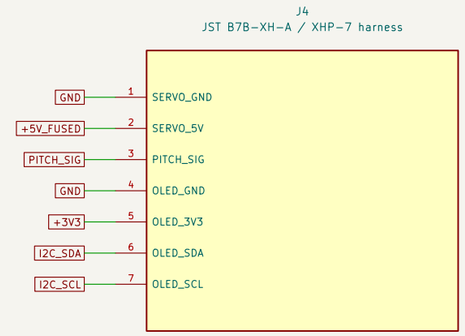








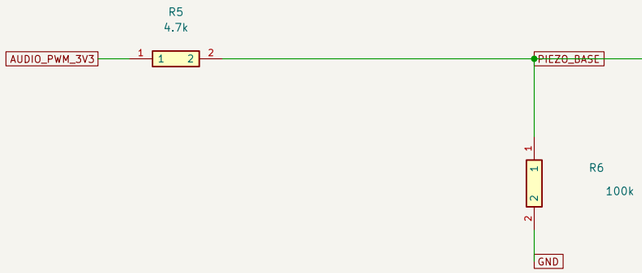


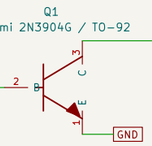


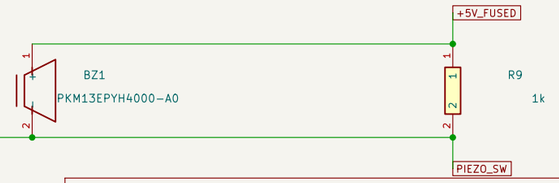
Do not feed +5V servo power through Pico, QT cable, or regulator.

J4 is the fixed-base disconnect. Seven wires form one loose rear service loop;

split into J3 three-wire servo lead and U3 QT1 four-wire display lead in head.







R9 and BZ1 are parallel. Q1 sinks the piezo negative terminal; emitter always GND.

All module external pins shown; vendor PCB internals referenced separately
SYSTEM SCHEMATIC / PROTOTYPE / HARDWARE UNTESTED
NODI — David Borsky

Sheet: /Actuators, buffer and piezo driver/
File: Motion_Audio.kicad_sch

Title: NODI / actuators and sound

Size: A3	Date: 2026-10-25	**Rev: B**
KiCad E.D.A. 10.0.7		Id: 3/3